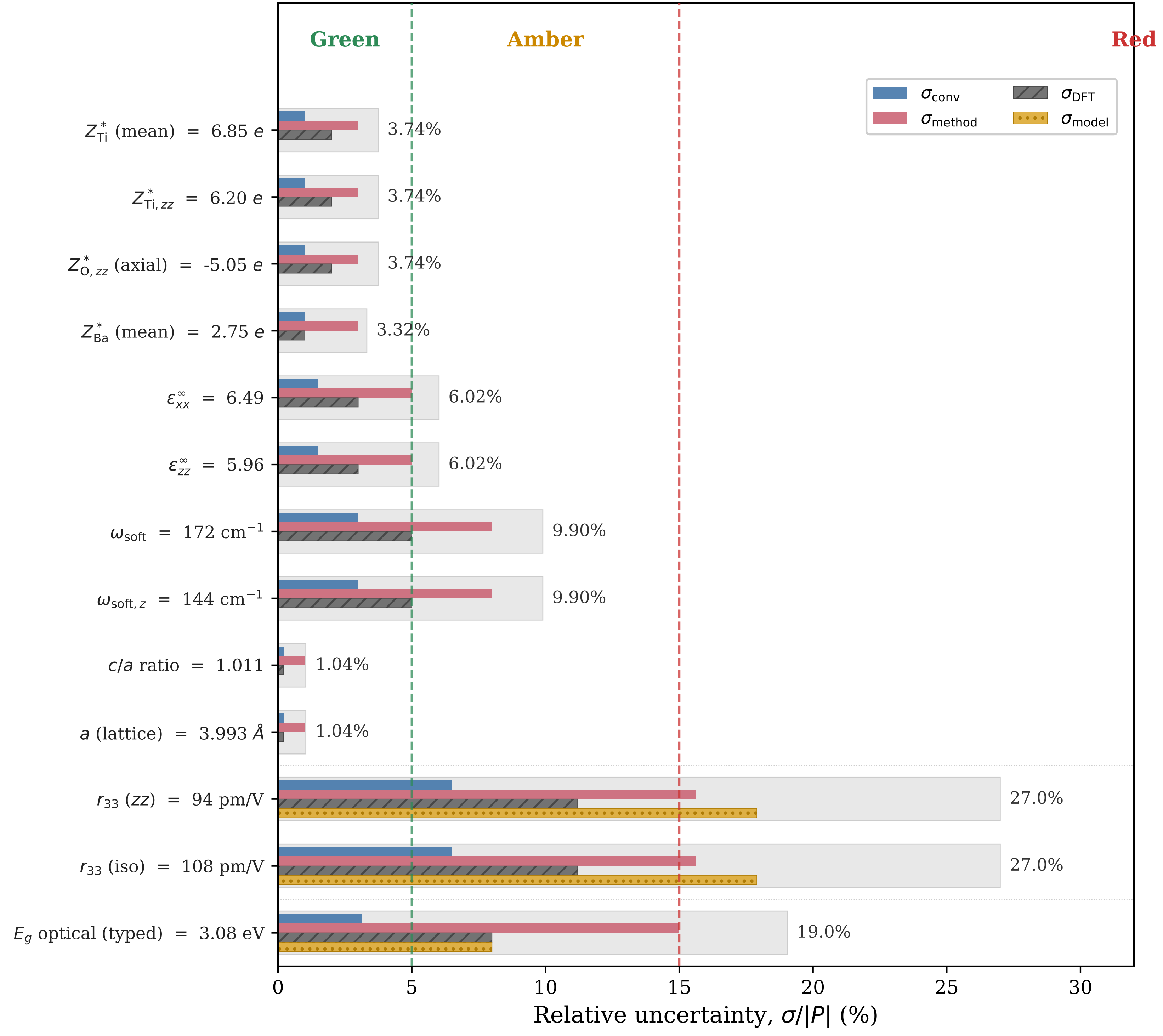
a) Certificate uncertainty decomposition



b) Typed band-gap JSON certificate


```
{
  "parameter": "E_gap_optical_typed",
  "value": 3.075,
  "unit": "eV",
  "target_observable": "room-temperature optical band gap",
  "quality_tier": "Red",
  "uncertainties": {
    "convergence": 3.125,
    "methodological": 15.0,
    "intrinsic_dft": 8.0,
    "model_form": 8.0,
    "total": 19.046
  },
  "non_exchangeability_rule": "Semilocal Kohn-Sham gaps are retained as Red-tier evidence but excluded from optical-gap fusion unless a calibrated reduction map is supplied.",
  "provenance": {
    "phase": "tetragonal",
    "physical_evidence": [
      "TB_mBJ_ACSOmega2020",
      "HSE06_proxy_Bilc2008",
      "optical_Wemple1970"
    ],
    "excluded_evidence": [
      "KS_LDA_Wahl2008",
      "KS_PBE_Wahl2008"
    ],
    "calibration_status": "prior HSE06 exact-exchange calibration should be imported before using calibrated-hybrid values as higher-tier priors"
  },
  "schema_version": "posterior-certificate-v1"
}
```


All uncertainty components are relative percentages.
The band-gap record demonstrates target-observable typing.